End-to-End Object Detection with Transformers: Fine-tuning DETR on Custom Datasets

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***Abstract*—This paper presents an implementation and analysis of the Detection Transformer (DETR), a novel approach to object detection that eliminates hand-designed components such as non-maximum suppression and anchor generation. We demonstrate fine-tuning DETR on a custom balloon dataset and provide insights into its transformer-based architecture, bipartite matching loss mechanism, and object query learning paradigm. The model achieves competitive performance while maintaining architectural simplicity and end-to-end trainability.**

***Index Terms*—Object detection, transformers, DETR, bipartite matching, computer vision, deep learning**

I. Introduction

Object detection has traditionally relied on hand-crafted components including anchor boxes, region proposals, and non-maximum suppression. The Detection Transformer (DETR) [1] reformulates object detection as a direct set prediction problem, streamlining the traditional pipeline through an end-to-end learnable architecture.

DETR demonstrates accuracy and runtime performance comparable to Faster R-CNN on COCO while offering conceptual simplicity and requiring no specialized libraries. This work presents an analysis of DETR's architecture and demonstrates its fine-tuning on a smaller, custom balloon detection dataset.

II. Methodology

A. Overall Architecture

DETR comprises three components: a CNN backbone for feature extraction, a transformer encoder-decoder for reasoning about object relationships, and prediction heads for generating detection outputs.

An input RGB image ($H_0 \times W_0 \times 3$) is processed through the CNN backbone to generate lower-dimensional features ($H \times W \times C$), typically with $C = 2048$, $W = W_0/32$, and $H = H_0/32$. For a $640 \times 640 \times 3$ input, the backbone produces a $20 \times 20 \times 2048$ activation map.

B. Transformer Encoder-Decoder

The backbone features are flattened and concatenated with spatial positional encodings for transformer processing. The encoder follows the original transformer design [2], with spatial encodings added only to Key and Query matrices.

The decoder introduces object queries—$N = 100$ learnable embeddings that produce bounding box predictions. Unlike anchor boxes encoding size priors, object queries learn spatial location patterns. Through training, each query specializes in detecting objects within specific image regions. Fig. 1 shows the spatial distribution of learned object queries.

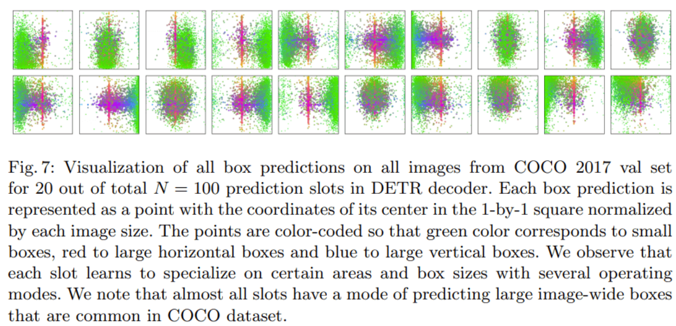

Fig. 1. Visualization of learned object query positions showing spatial specialization across the image domain.

C. Prediction Heads and Loss Function

Two feed-forward networks generate class predictions and bounding box coordinates for each query. When fewer than N objects exist, remaining predictions output a "no object" class.

DETR employs bipartite matching loss using the Hungarian algorithm to find optimal one-to-one assignments between predictions and ground truth (Fig. 2). The total loss combines classification and localization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{class}} + \lambda_{\text{L1}}\mathcal{L}_{\text{L1}} + \lambda_{\text{GIoU}}\mathcal{L}_{\text{GIoU}} \tag{1}$$

$$\hat{\sigma} = \arg\min_{\sigma \in S_N} \sum_{i=1}^{N} L_i(y_i, \hat{y}_{\sigma(i)})$$

Fig. 2. Bipartite matching between ground truth and predictions using the Hungarian algorithm.

III. Experimental Setup

We evaluate DETR on a custom balloon dataset with 610 training and 54 validation images. Annotations were converted from VIA to COCO format. The model was fine-tuned for 200 epochs using pretrained DETR weights, with the classification head retrained for single-class detection.

IV. Results and Discussion

A. Training Metrics

Average Precision showed consistent improvement over training (Fig. 3). Loss components including classification, L1, and GIoU losses demonstrated convergence (Fig. 4). Cardinality and class errors decreased throughout training (Fig. 5).

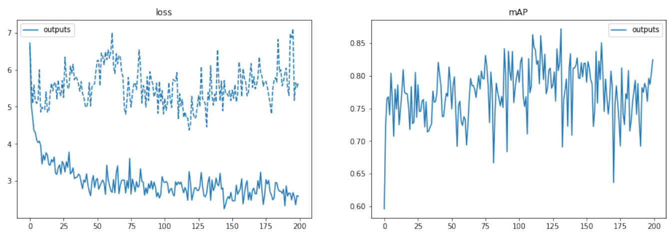

Fig. 3. Average Precision metrics over training epochs.

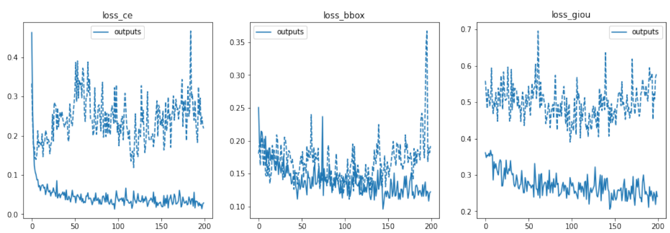

Fig. 4. Training and validation loss convergence.

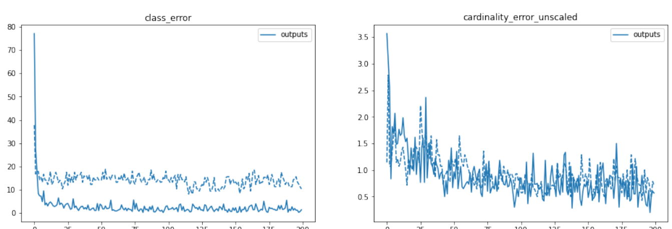

Fig. 5. Cardinality and class error metrics over training.

B. Qualitative Results

The model accurately detected balloon instances with precise localization, handling varying sizes, orientations, and partial occlusions without duplicate detections (Figs. 6, 7).

C. Key Advantages

DETR offers: (1) end-to-end training without separate stages, (2) elimination of anchors and NMS, (3) set-based prediction handling variable object counts, (4) global reasoning via transformer attention, and (5) architectural simplicity without specialized libraries.

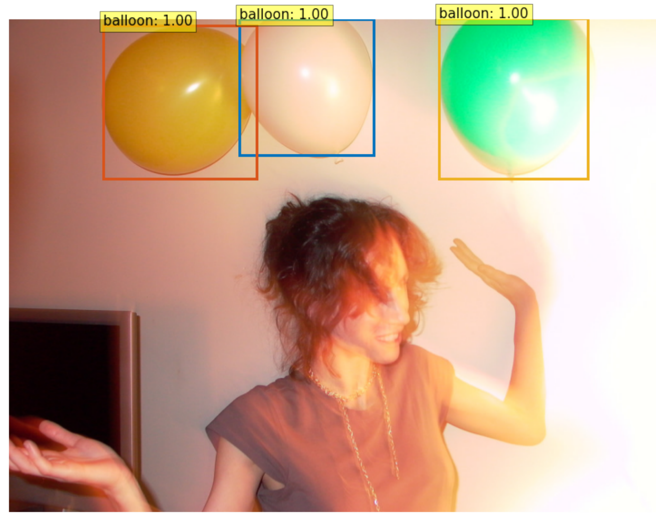

Fig. 6. Detection results showing accurate balloon localization.

Fig. 7. Additional results demonstrating robust detection across scales.

D. Comparison with Original DETR

My implementation differs from the original DETR in scale and approach. The original used 118,000 COCO images across 80 classes; we used 600 images for single-class detection. We employed transfer learning from pretrained COCO weights rather than training from scratch. Training used 200 epochs with simplified augmentation, achieving faster convergence due to pretrained initialization

V. Conclusion

This work demonstrated DETR's effectiveness through fine-tuning on a custom balloon dataset. The architecture's key innovations—object queries, bipartite matching loss, and end-to-end processing—provide a simpler alternative to traditional pipelines. Our comparison demonstrates transfer learning viability for specialized detection with limited data.

References

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